0625 | PAPER 4

TOPIC 04
Electricity and Magnetism

102 topical answers • 2021–2025
Calculations, final values and examiner-keyword summaries

Answers are independently condensed from the official marking requirements. They are not a reproduced Cambridge mark scheme.
For drawings and graph lines, compare the stated method with the original question figure.

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Core formulas

- charge $Q = It$
- potential difference $V = E \div Q$
- resistance $R = V \div I$
- power $P = IV = I^2R = V^2/R$
- energy $E = Pt$
- transformer: $V_p/V_s = N_p/N_s$
- ideal transformer: $V_pI_p = V_sI_s$

4.1 Current, charge and potential difference

Q186 • Oct/Nov 2021 Paper 41 Question 7 — 4.1 Current, charge and potential difference

(a) electrons mentioned
negative charges / electrons move from cloth or move to rod
(b)(i) electrons / negative charge(s) repelled to earth or ball charged by induction
ball positively charged
opposite charges attract
(b)(ii)
negatively charged (by rod) or ball discharges / becomes neutral
repelled by rod or pulled down by gravity / its weight

Q187 • Oct/Nov 2021 Paper 42 Question 8 — 4.1 Current, charge and potential difference

(a)(i) clearly more -ve (than +ve) on left and more +ve (than -ve) on right
same number of +ve and - ve
(a)(ii) -ve charges (flow) from earth or -ve charges flow to object
electrons flow to balance (excess) +ve charge on the object
(b) I = Q / t in any form or (Q =) It
(Q =) 0.65 × 10-3 × 2.2 × 60
(Q =) 0.086 C

Q188 • May/June 2022 Paper 41 Question 5 — 4.1 Current, charge and potential difference

(a) temperature
at which liquid becomes a gas or liquid and gas exist together
(b)(i) 1.8 × 105 J
(E =) VIt (in any form) or 230 × 13 × 60 or 230 × 13 or 3000
(b)(ii) 9.1 × 10-3 kg / s 9.1 × 10-3 kg / s
(ΔT =) 100 - 22 or 78 or (ΔT =) 100 - 22 or 78
m = E / cΔT (in any form) or 1.8 × 105 / (4200 × 78) or (rate =) P / cΔT (in any form) or m = E / cΔT and E = Pt
1.8 × 105 / (4200 × 78 × 60) or or 3000 / (4200 × 78)
.5 × 10N or 9.1/9.2 × 10N or 230 × 13 / (4200 × 78)
or 9.1 / 9.2 × 10N
(c) 1 if the tap becomes live or if the (live) cable touches the (metal) tap
there is a current to earth / in the earth wire (which blows the fuse)
2 the current (in earth wire) is large and fuse melts / blows / stops current / breaks circuit

Q189 • May/June 2022 Paper 43 Question 1 — 4.1 Current, charge and potential difference

(a)(i) 9.7 s
(a =) Δv ÷ t in any form or 28 (-0)/2.9
(a)(ii) 4600 N
(F =) ma in any form or 1600 × 2.9
(a)(iii) 630 000 J / 6.3 × 105 J
600 × 282
(KE =) ½ mv 2 in any form or
(b) 960 000 C / 9.6 × 105 C
(Q =) It in any form or 32 × 8.3 × 60 × 60
(t s =) 8.3 × 60 × 60
(c) any one explicit example of a variation from ideal conditions such as:
(repeated) acceleration / deceleration / use of brakes / varying speed
motion uphill / uneven road surface
cold weather / headwind

Q190 • May/June 2022 Paper 43 Question 2 4.1 Current, charge and potential difference

(a) gravitational potential (energy)

(b)(i) 2.0 × 106 J / s

(P =) E/t in any form or (480 × 10 × 410)/1

(ΔGPE =) mgh in any form or 480 × 10 × 410

(b)(ii) 81 (%) or 82 (%)

P = V I in any form or 6000 × 270 or 1 620 000

(efficiency =) (useful) power out / (total) power in (× 100%) in any form

(c)(i) damage to habitats (for fish) / construction is expensive / droughts / flood risk if dam bursts

(c)(ii) biofuel / wind / geothermal / tidal / solar / wave

Q191 • Feb/March 2023 Paper 42 Question 1 4.1 Current, charge and potential difference

(a)

speed = 4.3 m / s speed = 4.3 m / s

correct vector triangle or rectangle drawn use of Pythagoras' theorem

(

e.g. a2 + b2 = c2 or (speed =) √ 2.52 + 3.52)

direction = 54° or 55° direction = 54° or 55°

resultant velocity vector (including arrow) use of trigonometry to find angle

e.g. tan θ = 3.5 / 2.5

(b) a scalar quantity

distance, time, mass, energy, temperature

a vector quantity

force, weight, acceleration, momentum, electric field strength, gravitational field strength

Q192 • Oct/Nov 2023 Paper 41 Question 7 4.1 Current, charge and potential difference

(a) electrons move from cloth to rod

(plastic) rod gains electrons

(b)(i) (region) where an (electric) charge experiences a force

(b)(ii) At least three radial field lines distributed evenly around outside of S and touching S and not inside S

arrow on (at least one) field line pointing towards S

(c) arrow through Z and away from (centre of) sphere

Q193 • Oct/Nov 2023 Paper 43 Question 7 4.1 Current, charge and potential difference

(a)(i) radial arrow

inward radial arrow

(a)(ii) positive point charge

positive (charge)

(b) electrons / negative charges (move)

out of (plastic)

or removed from / lost from (plastic)

(c) any mention of free / mobile / delocalised electrons

conductors have free / mobile / delocalised electrons

insulators do not have free / mobile / delocalised

electrons

Q194 • May/June 2024 Paper 41 Question 6 4.1 Current, charge and potential difference

(a)(i) (region) where (an electric) charge experiences a force

(region) where a force acts on a (an electric) charge

(a)(ii) at least four straight radial lines and evenly spaced (by eye) surrounding sphere

four lines touching sphere and no lines inside sphere
at least one arrowhead towards sphere and no incorrect arrowheads

(b)(i) electrons move (through the wire) from the sphere

electrons move (through the wire) to(wards) the Earth
electrons move (in the wire)

(b)(ii) 2.5 A

I = Q / t or (I =) Q / t or 3.5 × 10-10 / 1.4 × 10-10
2.5 × 10N

Q195 • May/June 2024 Paper 43 Question 7

4.1 Current, charge and potential difference

(a)(i) evenly spaced straight vertical lines from one plate to the other

arrows on lines pointing towards negatively charged plate

(a)(ii) (negatively charged particle has) force / attraction towards positively charged / positive (plate)

(b)(i) 34 C

I = Q / t or (Q =) It or 28 000 × 0.0012

(b)(ii) (p.d. =) 3.6 MV or 3.6 × 106 V

E = ItV or (V =) E / It or (V =) E / Q or (V =) 1.2 × 108 / 34

Q196 • Oct/Nov 2024 Paper 43 Question 5

4.1 Current, charge and potential difference

(a)(i) four evenly spaced radial lines touching S and no lines inside S

at least one arrow pointing towards S and none incorrect

(a)(ii) to the left or away from the sphere / S

(a)(iii) accelerates (due to a resultant force)

(moves) away from the (centre of the) sphere

(b)(i) like charges repel

(b)(ii) advantage: more even coat; or less chance of paint particles clumping together

Key physics: explain positive / paint particles attracted same amount all parts negative sphere so spread out
or paint creates a fine mist so spreads out (over the surface)
or advantage: more durable; or better adhesion
less paint sag or less overspray on floor / walls
explain: the positive/paint is attracted to the negative sphere / charges and forms a strong(er) bond
or advantage: quicker to paint
Key physics: explain all surface painted same time without having turn sphere / go around other side

Q197 • May/June 2025 Paper 41 Question 4

4.1 Current, charge and potential difference

(a)(i) convection

(a)(ii) warm air rises or less dense air rises

warm air is less dense (than cool air)
Any one valid point(s):
- cold air replaces warm air
- cold air falls and the process repeats
- there is a convection current

(b)(i) P = IV or (I =) P ÷V

2.0 kW = 2000 W or 2000 ÷ 230

(b)(ii) 10 (A) and

Any one valid point(s):
- smaller fuse melts in normal use (of the heater)
- smaller fuse stops the heater working (at all)
- larger fuse allows too much current (without melting)
- larger fuse may not melt before the circuit is damaged
- fuse (rating) must be higher than the (normal) current
- the fuse will melt if current goes too high

Any one valid point(s):
- 10 (A)
- 13 (A) and fuse (rating) must be higher than (normal) current
- 13 (A) and 3 A / 5 A fuse melts in normal use

4.2 Electric circuits and resistance

Q198 • Feb/March 2021 Paper 42 Question 8
4.2 Electric circuits and resistance

(a) energy supplied by a source in driving charge around a complete circuit / energy needed to drive unit charge / 1 coulomb
round circuit
(b)(i) (P=) 90 W
(P=) V I in any form
(V / R or I =) 2
(b)(ii) (p.d. =) 15 V
(p.d. =) 60 - 45
(b)(iii) (I = 15 / 10 =) 1.5 A
(I =) V / R or V = IR in any form

Q199 • May/June 2021 Paper 41 Question 8
4.2 Electric circuits and resistance

(a)
and between P and Q
(b) 1.5 V c.a.o.
(c)(i) 1600 Ω
(V800 Ω =) 4.0 (V)
(I =) V / R in any form or 4.0 / 800 or 0.0050 (A) or (R =) V / I or 8.0 / 0.0050
1600 Ω
(V800 Ω =) 4.0 (V)
(RTh =) R800 Ω × VTh / V800 Ω in any form or (RTh =) 800 × 8.0 / 4.0 in any form
1600 Ω
12 8.0 RTh
or or
00+RTh RTh 800+RTh
12 8.0
= in any form
00+RTh RTh
(c)(ii) larger proportion of the e.m.f. (across thermistor) or smaller voltage across 800 Ω
temperature (of thermistor) is smaller / has decreased
resistance of thermistor / circuit is large(r)

Q200 • May/June 2021 Paper 43 Question 7
4.2 Electric circuits and resistance

(a) energy supplied
to drive a unit charge / 1 C round a complete circuit
(b)(i) (R =) 2.3 Ω or 2.2 Ω
R = V/I in any form
current in R = 4 (A) or p.d. across R = 9 (V)
(b)(ii) 1.1 Ω
resistance proportional to length (so twice length twice resistance)
resistance inversely proportional to area (so twice diameter decreases resistance by factor of 4)

Q201 • Oct/Nov 2021 Paper 41 Question 8
4.2 Electric circuits and resistance

(a) Q / t or (rate of) flow of (electric) charge / electrons

(b) I2 -

(current in the 450 Ω resistor =) ....

(c) (V450 Ω =) IR or 0.012 × 450 or 5.4 (V) or 9.0 - 5.4 or 3.6 (V) seen

(I =) 3.6 / 800 or 0.0045 (A)

(P =) VI or 3.6 × 0.0045 or 3.62 / 800

1.6 × 10-2 W or 16 mW

(d)

resistance (of LDR) decreases

current (in circuit) increases or resistance of parallel pair decreases

p.d. across 800 Ω resistor increases and or resistance of parallel pair a smaller fraction

p.d. across of total resistance and p.d.

450 Ω resistor decreases across 450 Ω resistor decreases

Q202 • Oct/Nov 2021 Paper 42 Question 9 — 4.2 Electric circuits and resistance

(a) 7.5 V

(b)(i) 1 / Rp = 1 / R1 + 1 / R2 or (Rp =) R1R2 / in any form

(Rp =) 1.2 (Ω)

3.2 Ω

(b)(ii) (V =) IR in any form

4.1 V

Q203 • Oct/Nov 2021 Paper 43 Question 8 — 4.2 Electric circuits and resistance

(a)

five straight, parallel vertical lines, equally spaced by eye, between plates

arrow head pointing upwards on at least one line and none wrong

(b)(i) 11 A

(I =) P / V in any form or 2400 = I220

(b)(ii) 9900 C or 9800 C

(Q =) It in any form or (Q =) 11 × 15 × 60

(b)(iii) 13 A

Q204 • Oct/Nov 2021 Paper 43 Question 9 — 4.2 Electric circuits and resistance

(a)(i)

four components joined in series

all circuit symbols correct for resistor, thermistor, a filament lamp and a power supply

(a)(ii) voltmeter connected in parallel to the resistor

(a)(iii) (p.d. across terminals of power supply) = 18 V

(current through resistor when p.d. across it is 6.0 V =) 0.4 A

current same through all components in series circuit

or horizontal line through 0.4 A on graph through all three curves

or p.d. across filament lamp = 3.0 V

or p.d. across thermistor = 9.0 V

p.d. across filament lamp = 3.0 V and p.d, across thermistor = 9.0 V

(b) any sensible use requiring temperature control; or depending on temperature, e.g. fire alarms, to keep computers cool (by

operating fan), in incubators, electronic thermometer, electronic thermostat in kettle / car engine

Q205 • Feb/March 2022 Paper 42 Question 7 — 4.2 Electric circuits and resistance

(a) 8 (cells)

(b)(i) (12 / 2.4 =) 5.0 Ω

R = V / I in any form

(b)(ii) (Resistance of Q = 5-1.5) = 3.5 Ω

1 / R = 1 / R1 + 1 / R2 or R= R1R2 /

(c)(i) correct voltmeter symbol connected correctly across both lamps

(c)(ii) 4.3 W

(P =) IV in any form or 1.2 × 3.6

3.6 V or 1.2 A

Q206 • May/June 2022 Paper 41 Question 6

4.2 Electric circuits and resistance

(a) Any three valid point(s):

radiation

light / infrared / electromagnetic (radiation)

travel through space / vacuum

absorbed by road

(b) road / black surfaces are good absorbers (of radiation); or sea is a poor absorber (of radiation)

(c)(i) they / molecules speed up or gain kinetic energy

they / molecules move further apart

(c)(ii) density (of air above road) decreases or density (of hot air) decreases

air (above land / road) rises or air (that is hot) rises

air (above road) replaced by cool air / air from above sea

Q207 • May/June 2022 Paper 43 Question 8

4.2 Electric circuits and resistance

(a)(i) light-dependent resistor / LDR

(a)(ii) voltmeter connected in parallel with component Y

(a)(iii) 1 0.016 A

(I =) V/R in any form or 12/400 or 12/350 or 12/750

(Rtotal = R1 + R2 =) 750 (Ω)

2 6.4 V

(a)(iv) (in a dark room the p.d. across component Y) decreases

(b) one named practical application of LDR e.g. switch on street lights (at night) / turn on security light (at night)

Q208 • Oct/Nov 2022 Paper 41 Question 2

4.2 Electric circuits and resistance

(a)(i)

magnitude or size

direction

(a)(ii)

Any two valid point(s): acceleration / deceleration, gravitational field strength, impulse, momentum, velocity, weight

(b)(i) 0.12 m

(b)(ii)

beyond where the extension is not directly proportional to the load; or (point) where extension stops being directly proportional to the load; or point up to which extension is directly proportional to the load

10.4 N ≤ weight ≤ 10.9 N

(b)(iii) 22 N / m ≤ k ≤ 25 N / m

clear subtraction of 0.12 from a length that is in Hooke's law region

e.g. 0.54 - 0.12

k = F / x in any form or k = W / x in any form or k = 1 / gradient

Q209 • Oct/Nov 2022 Paper 41 Question 8

4.2 Electric circuits and resistance

(a) both relate to energy per unit charge

(b) e.m.f. applies to the whole circuit / source; or p.d. to one (or more) component

from electrical for p.d.

(c)(i) 4.8 V

(c)(ii) 20 Ω
1 / RT = 1 / R1 + 1 / R2 or (RT =) R1 R2 / or 1 / RT = 1 / 24 + 1 / 12
or 1 / RT = 3 / 24 or (RT =) 24 × 12 / (24 + 12)
.0 (Ω)
(c)(iii) 2.9 V
V = ER / RT in any form or 4.8 × 12 / 20 or I = E / R in any form or 0.24 seen

Q210 • Oct/Nov 2022 Paper 42 Question 8 — 4.2 Electric circuits and resistance

(a) (RY) decreases
change in V consistent with stated effect on RY
change in RY / Rtotal consistent with their stated effect on RY
Key physics: change proportion total p.d. across Y fixed resistor consistent stated effect on RY
(b)(i) (n =) 1.9 × 1019
I = Q/t
(n =) 3(.0) / 1.6 × 10-19 or (n =) Q / 1.6 × 10-19
(b)(ii) (P =) 36 W
P = IV or (P =) IV or 3(.0) × 12

Q211 • Oct/Nov 2022 Paper 43 Question 8 — 4.2 Electric circuits and resistance

(a) positively charged / plastic rod is brought close to metal plate
negative charges / electrons (from metal plate) move to top of metal plate / close(r) to rod
earth lead connected to (metal) plate and negative charges / electrons move on to plate
(at the end of the process) earth lead removed (before charged rod removed)
or (at the end of the process) metal plate has (net) negative charge
(b) correct direction - pointing away from negative terminal / clockwise arrow
and current flow in opposite direction to flow of electrons

Q212 • May/June 2023 Paper 41 Question 4 — 4.2 Electric circuits and resistance

(a)(i) c = (Δ)E / mΔθ or (ΔE =) mcΔθ
(Δθ =) 21.5 - 19 or (Δθ =) 2.5 (°C)
(ΔE =) 0.6(0) × 4200 × 2.5 or (ΔE =) 0.6(0) × 4200 × {21.5 - 19}
(a)(ii) (maximum possible efficiency =) 3.1% or 0.031
E = Pt or (E =) Pt or (E =) 13 × 500 or (E =) 6500
(useful energy output =) 6500 - 6300 or (useful energy output =) 200
efficiency = useful energy (output) / total energy (input) (× 100%)
or (efficiency =) useful energy (output) / total energy (input) (× 100%)
or (efficiency =) {6500 - 6300} / 6500
or (efficiency =) 200 / 6500 (× 100%)
P = E/t or (P =) E / t or (P =) 6 300 / 500 or (P =) 12.6 (W)
(useful power output =) total power (output) - wasted power (output)
or (useful power output =) 13 - {6300 / 500}
or (useful power output =) 13-12.6
efficiency = useful power (output) / total power (input) (× 100%)
or (efficiency =) useful power (output) / total power (input) (× 100%)
or (efficiency =) 0.4 / 13 (× 100%)
(b) Any one valid point(s):
- temperature change is an underestimate (due to thermal energy losses)
- (thermal energy is) transferred from the water (to air / beaker / bench)
- energy (other than light) transferred in lamp (filament / glass / internal structure)
- (some) water evaporates

Q213 • May/June 2023 Paper 41 Question 7 — 4.2 Electric circuits and resistance

(a) Any two valid point(s):

• Key physics: potential divider splits / shares divides e.m.f. voltage difference p.d. power source circuit
• (e.m.f. is) split between (two) resistors / components (connected in series to power source)
• (potential divider shares e.m.f.) in proportion to the resistances (of the resistors / components)

(b)(i) (e.m.f. =) 15 V

(b)(ii) (resistance =) 60 Ω

(Rll =) R2R3 / or (Rll =) 40 × 40 / (40 + 40) or (Rll =) 1600 / 80
or 1 / Rll = 1 / R2 + 1 / R3 or 1 / Rll = 1 / 40 + 1 / 40 or (Rll =) (1 / 40 + 1 / 40)-1 or (Rll =) 20 (Ω)
(resistance =) 40 + (candidate's value for combined resistance of R2 and R3)

(c) (reading =) 10 V

emf shared in same proportion as resistance
or e.g. R1 / Rll = V1 / Vll
or (reading =) 15 × 40 / 60 or (reading =) 0.25 × 40

Q214 • May/June 2023 Paper 43 Question 6 — 4.2 Electric circuits and resistance

(a) (I =) 9.8 A

P
P = I 2R or I 2 = or I 2 =
R
I2 = 2500 / 26 or (I =) √(P / R) or

(b) (R =) 53 Ω

R ∝ l or 1.8 / 1.2 or 1.5 seen as multiplier
R ∝ 1 / A or 7.9 (× 10-7) / 5.8 (× 10-7) or 1.36(2) seen as multiplier

(c) (cost =) $36

E = Pt and (cost =) E × 0.3(0) or (cost =) 25 × 10N × 2 × 24 × 0.3(0)
or 3.6 × 10N dollars

Q215 • Oct/Nov 2023 Paper 41 Question 8 — 4.2 Electric circuits and resistance

(a)(i) 900 C

I = Q / t or (Q =) It or 1.5 × 600

(a)(ii) 2.0 Ω

R = V / I or (Rtot =) V / I or 9.(0) / 1.5 or 6.(0)
(Rcyl =) total resistance - P or (Rcyl =) 6.0 - 4.0

(b) 1200 s

R is directly proportional to l or (new cylinder) twice as long means twice R
R is inversely proportional to A or (new cylinder) half cross-sectional area means twice R
(resistance of cylinder =) 4 × (a)(ii) (Ω)

Q216 • Oct/Nov 2023 Paper 43 Question 6 — 4.2 Electric circuits and resistance

(a) sketch:

Key physics: axis labelled V other I either way round gradual curve origin curving such /
increases
explanation:
(as current / temperature increases so) resistance increases
(so) more voltage required for same increase in current
or if V on y-axis: gradient must increase
or if V on x-axis: gradient must decrease

(b)(i) 9.0 V

12.0 - 3.0

(b)(ii) 2.1 A

I = V / R or (I =) V / R or 9.0 / 4.2

(b)(iii) 0 (A)

(b)(iv) (ammeter reading) 2.1 A

(b)(v) 19 W

P = VI or or (P =) VI or 9 × 2.1

Q217 • Feb/March 2024 Paper 42 Question 7 — 4.2 Electric circuits and resistance

(a)

(b)(i) 1.5 V

Vout / VR = Rout / R or VR = 3 Vout

(b)(ii) 0.51 C

I = Q / t or (Q =) It or 1.7 × 10-3 × 300

Q218 • May/June 2024 Paper 41 Question 7 — 4.2 Electric circuits and resistance

(a) (electrical) work done moving a unit charge around a (complete) circuit

work done and moving a charge (in a circuit)

(b)(i) correct symbols for five cells in series

correct symbols for variable resistor and fixed resistor

cells, variable resistor and fixed resistor connected in series

(b)(ii) curve with negative gradient of decreasing magnitude from 0 Ω to 150 Ω and does not reach the x-axis

curve / line with negative gradient from 0 Ω to 150 Ω

y-axis labelled 0.25 where candidate's line meets the y-axis or the mark on the y-axis labelled 0.25

R = V / I or (Imax =) V / R or 7.5 / 30

Q219 • May/June 2024 Paper 42 Question 3 — 4.2 Electric circuits and resistance

(a) gravitational (potential) energy (store before / as the ball falls)

kinetic energy (store) increases (as the ball falls); or energy transferred to kinetic energy (store as the ball falls)

(energy transferred from) kinetic energy (store) to internal / thermal energy (store)

(b) 5.9 m / s

(Δ)Ep = (Δ)Ek or Ep lost = Ek gained or gravitational potential energy lost = kinetic energy gained or mg(Δ)h = ½mv2

v2 = 2g(Δ)h or v2 = 2 × 9.8 × 1.8 or v2 = 35(.28)

Q220 • May/June 2024 Paper 42 Question 8 — 4.2 Electric circuits and resistance

(a)(i) correct voltmeter symbol connected across LDR

(a)(ii) 1 (resistance) increases

(a)(ii) 2 (p.d.) increases because resistance of parallel combination of LDR and LED increases

greater proportion of (total) p.d. across LDR / LED / parallel combination of LDR and LED

(b) 960 Ω

current in each bulb = 0.25 or R = V / I or (R =) V / I

resistance = 240 / 0.25 or 1 / 480 = 1 / R + 1 / R

Q221 • May/June 2024 Paper 43 Question 1 — 4.2 Electric circuits and resistance

(a) (deceleration is) decrease in velocity per unit time; or rate of decrease in velocity

or -Δv / Δt

negative acceleration or change in velocity per unit time or rate of change of velocity

(b) a = Δv / (Δ)t and (Δt =) 14 / 9.8 or (Δt =) Δv / a = 14 / 9.8

(c) 10 m

(initial) Ek of ball = (maximum) Ep gained or ½ mv2 = mgh

(h =) Ep / mg or (h =) ½ v2 / g or (h =) 12.74 / (0.13 × 9.8) (= 10 m) or (h =) (½ × 196) / 9.8

(d) Any three valid point(s):

- (ball) accelerates
- (accelerates) at 9.8 m / s2 initially or (accelerates) due to force of gravity
- air resistance / resistive force increases (with speed / velocity)
- resultant force (downwards) decreases
- acceleration decreases
- terminal velocity is reached when acceleration is zero; or terminal velocity is reached when resultant force is zero

Q222 • May/June 2024 Paper 43 Question 2
4.2 Electric circuits and resistance

(a) chemical (energy store)
(b) if one lamp breaks, the other one will remain lit
(c)(i) the useful energy / power output from the solar panel is 22% of the total energy / power input
(c)(ii) 68 W
(%) efficiency = {useful power output} / {total power input} (× 100%) or 15 / 0.22
(d) Any two valid point(s):
- no need for cables (to connect to mains) / good for locations remote from mains supply
- less power loss than mains (electricity) that must be transmitted
- not affected by mains power cuts

Q223 • Oct/Nov 2024 Paper 41 Question 1
4.2 Electric circuits and resistance

(a)(i) 43 cm and 63 cm
(a)(ii) 20 cm
(b) 0.28 N / cm
k = F / x or (k =) F / x or 5.6 / 20
(c)(i) 4.9 N
(c)(ii) 3.2(0) m / s2
F = ma or (a =) F / m or (6.5 - 4.9) / 0.50
(resultant force =) 6.5 - 4.9 or 1.6

Q224 • Oct/Nov 2024 Paper 43 Question 6
4.2 Electric circuits and resistance

(a)(i) Any two valid point(s):
- if one resistor fails / breaks the others will still work
- each resistor gets the full voltage / 12 V or lower (total) resistance or higher current
- higher power

(a)(ii) Any two valid point(s):
- evaporation / water evaporates
- energy (from heater) transfers to particles or particles gain energy (from the heater)
- more energetic particles escape (from water / droplet)
- particles leave from the surface (of the water / droplet)

(b)(i) 0.17 A
R = V / I or (I =) V / R or (I =) 12 / 70
(b)(ii) 610 J or 620 J
E = I × V × t or (E =) V × I × t or (E =) 12 × 0.17 × 300
300 (s) or 5 × 60
(b)(iii) chemical
(b)(iv) 39 Ω
1 / RT = 1 / R1 + 1 / R2 or 1 / RT = 1 / 90 + 1 / 70 or (RT =) 1 / (1 / R1 + 1 / R2) or (RT =) 1 / (1 / 90 + 1 / 70)

Q225 • Feb/March 2025 Paper 42 Question 5
4.2 Electric circuits and resistance

(a) y-axis labelled resistance and x-axis labelled light intensity
Straight line / smooth curve with negative gradient
(b)(i)
Correct symbol drawn to complete circuit.

(b)(ii) in the dark, VLDR is a bigger proportion of e.m.f or
when RLDR is high VLDR is a bigger proportion of e.m.f. or
when VLDR is high VLDR is a bigger proportion of e.m.f.
In the dark, VLDR is high or
when RLDR high, VLDR is high
Any one valid point(s):
• emf shared (between fixed resistor and LDR) or emf is constant
• VLDR = VLAMP or p.d. is the same across components in parallel

Q226 • May/June 2025 Paper 41 Question 7 — 4.2 Electric circuits and resistance

(a)(i) diode
Key physics: only current diode voltage increased direction
(a)(ii)
braille: symbol A identified
(b)(i) 59 Ω
V = IR or (R =) V ÷ I or (R =) 230 ÷ 3.9
(b)(ii) 270 000 J or 2.7 × 105 J
(E =) IVt
(t =) 5 × 60 seen
(b)(iii) 7.8 A
Any one valid point(s):
• identical heaters (each with p.d. of 230 V) so 3.9 A in each branch
• I = I1 +
1 1 1
• = + in any form
R R1

Q227 • May/June 2025 Paper 42 Question 7 — 4.2 Electric circuits and resistance

(a) (region) where a(n electric) charge experiences a force
(region) where a force acts on a(n electric) charge
(b)(i) Any two valid point(s):
• friction (between cloth and rod causes electrons to gain energy)
• electrons move
• (electrons move) from cloth / to plastic (making plastic negative and cloth positive)
(b)(ii) moves away (from the charged plastic rod)
moves / experiences a force / repels
(c)(i) energy transferred in one hour at a rate of transfer of 1 kW
(c)(ii) 1 5.5 (kW h)
P = (Δ)E ÷ t or (ΔE =) Pt or (ΔE =) 0.025 × 220 or 25 × 220
2 0.11 × A
P = V I or (I =) P ÷ V or (I =) 25 ÷ 230

Q228 • May/June 2025 Paper 43 Question 7 — 4.2 Electric circuits and resistance

(a) 0.5(0) A
(b) 3(.0) V
(V =) IR or 0.5(0) × 6(.0)
(c) 12 - 3(.0)
(d) 180 Ω
I2 = 9.0 ÷ 20 or I2 = 0.45 (A) or I1 = 0.05 (A)
(R =) 9 ÷ 0.05

Q229 • May/June 2025 Paper 43 Question 8 — 4.2 Electric circuits and resistance

(a)(i) (region) where (an electric) charge experiences a force
or (region) where a force acts on a (an electric) charge
(a)(ii) four radial straight field lines starting at charge
at least one arrow on a field line towards the charge
(b) 1600 C
V = W / Q or (Q =) W / V
(Q =) 4.5 × 10N ÷ 2.9 × 10 8 or 1.6 × 10N
(c) 240 C
(Q =) It or 2(.0) × 120

Q230 • Oct/Nov 2025 Paper 41 Question 7 — 4.2 Electric circuits and resistance

(a) 38 Ω
(R =) V ÷ I or (R =) 3 ÷ 0.08(0)
(b)(i) Current in lower branch is 0.16(0) A
ammeter reading is sum of currents in each branch
(ammeter reading =) 0.24 A
resistance in top branch = twice value in 7(a)
formula for combined resistance of resistors in parallel
Correct current calculated from candidate's R1 in 7(a)
(b)(ii) potential difference (p.d.)(across R 3) is larger (than p.d. across R1)
more work done (passing charge through R3)

Q231 • Oct/Nov 2025 Paper 43 Question 6 — 4.2 Electric circuits and resistance

(a)(i) 0.008(0) A or 8(.0) mA or 8(.0) × 10-3 A
(p.d. across 400 Ω resistor =) 5 - 1.8 or 3.2 (V)
(I =) V / R or 3.2 / 400
(a)(ii) increases
resistance of LDR increases
ratio of potential differences (across each component) equals ratio of the two resistances or R1 / R2 = V1 / V2
LDR has greater share of the total / supply voltage
potential difference of the source is divided between the resistor (R) and LDR
(b)(i) 56 mA or 5.6 × 10-2 A
44 (mA) or 12 (mA)
(b)(ii) 0 (A)

4.3 AC and DC circuits

Q232 • May/June 2022 Paper 42 Question 8 — 4.3 AC and DC circuits

(a) (t = 1/60 =) 0.017 s or 1.7 × 10-2 s
(b)(i) diode
(b)(ii) (I =) 1.4 A
(I =) Q / t in any form
(I =) 1.5 × 1017 × 1.6 × 10-19 / 0.017 or 0.024 / 0.017
(c) one arrow clockwise and one arrow anticlockwise
arrow anticlockwise (around circuit) labelled I
(d) (P = 0.35 × 12 =) 4.2 W
(P =) IV in any form

Q233 • Oct/Nov 2022 Paper 43 Question 9 — 4.3 AC and DC circuits

(a) 0.02 s
t =1 / f or (t =)1 / f or 1 / 50
(b)(i) correct shape shown with rectification for two cycles
sine shape shown (without rectification for two cycles)
(b)(ii) 340 marked
(b)(iii) one correct time value marked on time axis
a second correct time value marked on time axis

Q234 • Feb/March 2023 Paper 42 Question 7 4.3 AC and DC circuits

(a) work done in passing charge through / across a component
work done per unit charge
(b)(i) (definition of emf:) E = W / Q
(b)(ii) 270 J
W = EQ or 9.0 × 30
(c)(i) correct symbols for d.c. power supply, a lamp and a thermistor
three components in a complete series circuit
(c)(ii) resistance (of thermistor) decreases (when temperature increases)
resistance of circuit decreases; or greater current (in lamp so brightness of lamp increases)
(so brightness of lamp increases)

Q235 • Oct/Nov 2024 Paper 41 Question 6 4.3 AC and DC circuits

(a) work done by a unit charge passing through a component
(electrical) work done and moving charge
(b) (p.d. =) E - reading on voltmeter or subtract reading on voltmeter from E
(c)(i) (intensity of light on LDR) increased
(temperature of thermistor) decreased
(c)(ii) reading on voltmeter / it decreases
Any two valid point(s):
1 e.m.f. is constant
2 RLDR / Rthermistor decreases or RLDR is a smaller proportion of the total resistance
3 VLDR / Vthermistor decreases or VLDR is a smaller proportion of e.m.f.
R1 V1
4 =
R2 V2

Q236 • Oct/Nov 2025 Paper 42 Question 3 4.3 AC and DC circuits

(a) more electricity generated or makes solar panels more efficient
black is a good absorber or black is a poor reflector
(b) 22 kW or 22 000 W
(total power input =) {useful power output (× 100%)} / (%) efficiency
or (total power input =) {3.5 × 100} / 16 or (total power input =) 3.5 / 0.16
(total power input =) 3.5 / 0.16 or (total power input =) {3.5 × 100} / 16
(c) alternating current reverses direction or direct current is only in one direction
(d) electricity can be used when there is no Sun; or when it is dark
(e) 29 000 C
(Q =) I t or (Q =) 4.0 × 2.0 × 60 × 60
correct conversion from h to s SEEN

Q237 • Oct/Nov 2025 Paper 42 Question 7 4.3 AC and DC circuits

(a) correct symbols for a.c supply and lamps
components all connected in parallel
(b)(i) 0.29 A

(I =)V / R or (I =)230 / 800

(b)(ii) 0.33 (kWh)

(E =)VIt or (E =) {230 × 0.29 × 5(.0)} / 1000

conversion from W to kW seen i.e. division by 1000

(b)(iii) 480 Ω

1 / R = 1 / R1 + 1 / R2 or 1 / R = 1 / 800 + 1 / 1200 or (R =) R1R2 / {R1 + R2} or

(R =) {800 × 1200} / {800 + 1200}

4.4 Magnetism and magnetic fields

Q238 • Feb/March 2021 Paper 42 Question 7

4.4 Magnetism and magnetic fields

(a) no cutting of (magnetic) flux / magnetic field

(b) to the top of the page / RH box

current, motion and (magnetic) field

mutually at right angles

(magnetic) field from left to right

(c)(i) opposite current (direction) / opposite deflection (on ammeter)

(c)(ii) greater current / deflection

Q239 • May/June 2021 Paper 42 Question 7

4.4 Magnetism and magnetic fields

(a) 3 lines from N face to S face

middle line must be straight and perpendicular to end faces

at least 1 arrow from N to S

and NO arrows from S to N

(b)(i) needle perpendicular to end faces and

{arrow pointing to S or correctly labelled N or S}

(b)(ii) compass / needle / it aligns with field or

compass / needle / it points in direction of magnetic field

or compass / needle / it points to S(outh)

N pole of needle attracted to S of magnet(s)

or N pole repelled by N of magnets

or unlike poles attract / like poles repel

(c) heat or hammer

with magnet lying (magnetically) E - W

or place in coil / solenoid with a.c.

withdraw or reduce current to 0

Q240 • May/June 2022 Paper 41 Question 8

4.4 Magnetism and magnetic fields

(a) 0.27 J

(v =) at (in any form) or 10 × 0.67 or 6.7 (m / s)

6.7 (m / s)

(KE =) ½mv 2 (in any form) or ½ × 0.012 × (10 × 0.67)2

or ½ × 0.012 × 6.72

(b)(i) magnetic field / magnetic field lines cut the copper / tube / it (or vv.)

electromagnetic induction occurs or e.m.f. induced

(b)(ii) (upwards / opposing) force on magnet

Key physics: force / magnetic field e.m.f. current opposes change producing motion magnet due

magnetic field caused by current in tube

Q241 • May/June 2022 Paper 42 Question 7

4.4 Magnetism and magnetic fields

(a) (pole A:) N and (pole B:) S

(b) vertical
up
(c) vertical
down

Q242 • Oct/Nov 2022 Paper 41 Question 5 4.4 Magnetism and magnetic fields

(a) infrared
(b)(i) (both) transverse / electromagnetic / travel in a vacuum / have the same (high) speed (in a vacuum)
(b)(ii) (it / visible light) compared with an e.m. radiation stated by candidate in 5(a) in terms of frequency / wavelength
(c)(i)
equipment
e.g.
black container, white container, thermometers
Leslie's cube and detector
measurements made
warm / hot water in container and temperature decreases recorded; or time to reach a given temperature / to cool
warm / hot water in cube and meter readings recorded
how a conclusion is reached
better emitter surface cools quicker
greater reading from better emitter surface
(c)(ii) any two appropriate quantities
e.g.
initial temperature of water
mass / volume of water
dimensions / surface area of container
time of cooling
mass of container
shape of container
smoothness of surface
surface area of face (of cube)
distance of detector
temperature of water at time
of measurement
smoothness of surface

Q243 • Oct/Nov 2022 Paper 42 Question 7 4.4 Magnetism and magnetic fields

(a) (minimum of) one complete loop above magnet and one complete loop below magnet
additional field lines leaving both poles or additional loops above and below
(minimum of) two correct arrows (from N to S)
(b) line with arrow to the left
(c)(i) (force to the) left or (force) away from magnet 2 / towards magnet 1
(c)(ii) force (on N pole) is in direction of the (magnetic) field /

Q244 • May/June 2023 Paper 42 Question 8 4.4 Magnetism and magnetic fields

(a)(i) region in which a (magnetic) pole experiences a force
(a)(ii) in the direction of the force on the N pole
(b) 4 radial lines outside sphere, touching the sphere and equally spaced all around sphere
direction of arrows towards the sphere
(c)(i) 2.0 V
(c)(ii) (ratio of p.d. across R2 : R3 =) 1 : 2
(c)(iii) 1. current is zero in R1 and diode is in wrong direction (to current)

(c)(iii) 2. (ratio of p.d. across R2 : R3 =) 1 : 1

Q245 • Oct/Nov 2023 Paper 42 Question 7 — 4.4 Magnetism and magnetic fields

(a)
all correct 2 marks
1 or 2 correct 1 mark
(b) 3.0 × 108 m / s
(c)(i) 0.12 m
(mid-point of frequency range identified as) 2.44 (GHz)
v = fλ or (λ =) v / f or (λ =) 3.0 × 108 / 2.44 × 109 or (λ =) 1.2 × 10N
(c)(ii) (radio waves / signal) lose energy / get weaker / lose (signal) strength (passing through walls)

Q246 • Feb/March 2024 Paper 42 Question 8 — 4.4 Magnetism and magnetic fields

(a)(i) three concentric circles centred on X
second and third circles further apart than first and second circles
direction of arrows clockwise
(a)(ii) strength of magnetic field increases
its direction reverses
(b) Any three valid point(s):
• P = I 2R or power (loss) = I 2R
• (high voltage allows) low current (so at same power output, less power / energy lost)
• thin wires have high resistance (so more power / energy lost)
• (low) current has a greater effect (on efficiency) than (high) resistance (of the thin wires)

Q247 • Feb/March 2025 Paper 42 Question 3 — 4.4 Magnetism and magnetic fields

(a) infrared
(b) shiny surface / it is a good reflector of radiation
Any one valid point(s):
• it is a good reflector
• it reflects radiation
(c) Any one valid point(s):
• prevents (electric) shock (if live wire touches the metal casing)
• if live wire touches the metal casing the current goes to earth
(d)(i) 2.6 A
R = V / I or (I=) V/R or (I=) 230 / 89
(d)(ii) P = IV
(I =) 5.2 (A) or (P =) 2 × power of one element or
(d)(iii) 68 000 J or 68 kJ
E = Pt or (E =) Pt or (E =) 1200 × 60
efficiency = useful energy out / total energy (in) or 95 ÷ 100 ×E
(power output of heater =) 95% × 1200

Q248 • May/June 2025 Paper 41 Question 6 — 4.4 Magnetism and magnetic fields

(a) Any one valid point(s):
• magnetic materials are attracted to magnets
• non-magnetic materials are not attracted to magnets
• magnetic materials experience a force in magnetic fields
• non-magnetic materials don't experience a force in magnetic fields
(b) Any one valid point(s):
• steel as it stays magnetised
• steel makes a permanent magnet
(c) N and S poles correctly labelled

braille: north to south

(d) field lines close(r) indicates strong(er) (magnetic) field

Q249 • Oct/Nov 2025 Paper 41 Question 8
4.4 Magnetism and magnetic fields

(a)(i) field lines parallel inside solenoid

Minimum of 2 magnetic field lines, curved towards end of solenoid, around outside of solenoid

Correct symmetrical pattern above and below solenoid and no field lines crossing and At least 2 complete lines

Key physics: least arrow field line outside solenoid away left end towards right

(a)(ii) B marked at a place where field lines are close together in Fig. 8.1 and (magnetic) field lines are close(r) together

(b)(i)

(b)(ii) Any one valid point(s):

• solenoid becomes an (electro)magnet

• there is a magnetic field (around solenoid)

Any one valid point(s):

• attraction between solenoid and the (soft) iron (arm)

• (soft) iron becomes an (induced) magnet

(b)(iii) Any one valid point(s):

• contacts are broken

• circuit is broken (when striker hits bell)

• there is no current (in the solenoid)

Any one valid point(s):

• solenoid stops being a magnet

• soft iron loses magnetism

• soft iron stops being attracted to solenoid

• Springy metal moves the soft iron back

4.5 Electromagnetic induction

Q250 • Feb/March 2022 Paper 42 Question 6
4.5 Electromagnetic induction

(a) Minimum of one arrow on a field line pointing N → S and not contradicted

central line perpendicular to poles of magnets and at least two other correct field lines

(b)(i) (A are) slip rings

(b)(ii)

Minimum of one arrow, pointing clockwise, on the wire, not contradicted

Field direction, motion of wire and induced current are mutually perpendicular

(b)(iii) (as coil rotates) it cuts (magnetic) field between the magnets

This induces an e.m.f. / voltage / p.d. (in the coil)

This produces a current in the (coil transferred to the) galvanometer (via the slip rings and carbon brushes)

Direction of current flow changes with each 180 degree rotation of coil

Q251 • May/June 2024 Paper 43 Question 9
4.5 Electromagnetic induction

(a) (alternating current / a.c. in primary coil / plate produces) changing magnetic field (in primary coil)

secondary / phone coil cuts (this) magnetic field; or secondary / phone coil is in this / changing magnetic field

(changing magnetic field) causes induced current (in secondary coil)

(b)(i) 1 kW h = 1000 × 60 × 60 J and 0.012 × 3.6 × 106 (= 4.3 × 104 J)

(b)(ii) 50% charged = (4.3 × 104 / 2 =) 2.15 × 104 (J) or 63% charged in 30 min

or (50% charged in t =) 24 min

P = E / t or (t =) E / P or 2.15 × 104 / 15 (s)

or energy provided in 30 min = (15 × 30 x 60=) 2.7 × 104 (J)

2.7 × 104 > 2.15 × 104 or 63% > 50% or 30 min > 24 min

Q252 • Oct/Nov 2024 Paper 41 Question 7 — 4.5 Electromagnetic induction

(a) (soft) iron

(b)(i) (at least) one complete field line between the poles of the bar (either above; or below the bar)

no crossing and attempt at correct shape and at least six lines from / to poles

at least one arrowhead towards S pole

(b)(ii) current (in the coil) decreases

(current decreases so magnetic field) strength decreases

(field strength decreases so) fewer field lines (in same area); or (field strength decreases so) field lines further apart

(c) Any two valid point(s):

1 (changing resistance causes) changing current (through solenoid)

2 (changing current causes) changing magnetic field (around solenoid)

3 (square) coil cuts (changing) magnetic field or coil in changing magnetic field

e.m.f. induced (between terminals)

Q253 • Oct/Nov 2024 Paper 43 Question 7 — 4.5 Electromagnetic induction

(a)(i) 100 (Nm)

(a)(ii) 0.44 m or 0.45 m

total clockwise moment = total anticlockwise moment

(d =) 100 / 225 or (d =) 100 / (23 × 9.8) or 100 / (23 × g × d)

(b)(i) (barrier arm) rotates anticlockwise or (RHS of barrier arm) moves upwards

(when the switch is closed) coil produces a magnetic field; or coil / core becomes an (electro)magnet

(coil/core/electromagnet) attracts (iron) bar (downwards)

(b)(ii) statement: (barrier arm) moves slower / goes up (more) slowly

explanation: decreases strength of (magnetic) field / smaller force / smaller moment

(b)(iii) steel would become permanently magnetised

(so when switch is opened) barrier would stay up; or bar A will stay attracted (to soft iron core)

Q254 • Oct/Nov 2025 Paper 42 Question 8 — 4.5 Electromagnetic induction

(a) region in which a (magnetic) pole experiences a force

(b)(i) friction between trolley and track; or the (induced) current produces a magnetic field which opposes the (magnetic) field

that causes the (induced) current

(b)(ii) Any three valid point(s):

- direction will (still) be left or there is no change of direction
- induced current / emf / voltage does not change direction
- smaller deflection
- smaller (rate of) change of magnetic field
- smaller current / emf / voltage induced (in coil)

4.6 Generators

Q255 • Oct/Nov 2021 Paper 43 Question 2 — 4.6 Generators

(a) (rate of transfer of gravitational potential energy =) 0.17 W

(gravitational PE lost =) mgh in any form or 12 × 10 × 1.7

(gravitational PE lost =) 204 (J)

(gravitational PE lost / s =) 204 / 1200

(b) 59% or 0.59

efficiency = useful power output / power input (× 100%) in any form or 0.10 / 0.17 × 100%

(c) any sensible advantage, e.g. no use of (fossil) fuel, no cost to run, can be used in remote areas, no CO_2 / air pollution, no

greenhouse gases, does not contribute to global warming

Q256 • Oct/Nov 2022 Paper 43 Question 3 — 4.6 Generators

(a) 5 correct: 3 marks, 3 or 4 correct: 2 marks, 2 correct: 1 mark
gravitational potential
water
(tidal) bay
kinetic
turbines
(b) any one advantage from:
• renewable
• reliable or predictable
• running cost low
• does not produce (harmful) pollution.
any one disadvantage from:
• (high) cost of construction
• possible effects on (marine) life
• not available all day
• power produced doesn't always match with peak demand
• limited number of sites
• maintenance difficult / increased corrosion (because underwater).
(c) Moon

Q257 • Oct/Nov 2023 Paper 43 Question 8 — 4.6 Generators

(a)(i) clockwise
force on left of coil up or force on right of coil down
(a)(ii) (turning effect is) greater
(a)(iii) (turning effect is) less
(b)(i) horizontal line, labelled B on the left and A on the right
(b)(ii) vertical line, labelled A at top and B at bottom
(b)(iii) horizontal line, A on the left and B on the right

Q258 • May/June 2025 Paper 42 Question 8 — 4.6 Generators

(a)(i) P and Q: slip rings
X and Y: brushes
(a)(ii) coil cuts magnetic field
(a)(iii) Any one valid point(s):
• increase strength of magnetic field
• increase speed of rotation of coil
• increase number of turns (of coil)
(b)(i) 1.2 A
(total) resistance of two (identical) resistors in series is added / doubled
(b)(ii) 6(.0) A
(b)(iii) 2.1 Ω
$V = I R$ or $(R =) V \div I$ or $(R =) 5(.0) \div 2.4$

Q259 • Oct/Nov 2025 Paper 43 Question 7 — 4.6 Generators

(a)(i) provide a continuous connection (between the coil and the lamp)
prevent the contacts reversing every half turn / 180°
prevent the wires (to the lamp) twisting / tangling
(a)(ii) Any three valid point(s):
• coil is rotated / moved in magnetic field
• coil cuts (magnetic) field or changing (magnetic) field through coil

• e.m.f. / voltage induced
• coil forms part of a complete circuit (through the lamp) so there is a current
• as one side of the coil starts to move in the opposite direction, current is induced in the opposite direction
or direction of current in coil is reversed every half turn / 180°

(a)(iii) Any two valid point(s):

• increase strength of magnetic field
• increase speed of rotation (of the coil)
• increase number of turns (of coil)

(b)(i) 2.5 (revolutions)

(b)(ii) any 1 correct = 1 mark

any 2 or 3 correct = 2 marks
all 4 correct = 3 marks
Time Label
coil position
f rom graph
A
N S
B N S
C
N S
D N S

4.7 Motors

Q260 • May/June 2021 Paper 42 Question 9 — 4.7 Motors

(a) anti-clockwise arrow labelled (conventional) current somewhere in circuit

electron (flow) arrow opposite to (conventional) current

(b) Q = It in any form or (Q =) It or 13 × 1

(Q = It =) 13 × 1 (= 13 C)
(n = 13 / 1.6 × 10-19 =) 8.1 × 1019

Q261 • May/June 2022 Paper 42 Question 1 — 4.7 Motors

(a)(i) (E =) 2 200 000 (J) or 2.2 × 106 (J)

(E =) Pt in any form
(E =) 600 × 3600

(a)(ii) chemical

(b) (t =) 8600 s or 140 min or 2.4 h or 2 h 24 min

(t =) 8800 s or 147 min or 2 h 27 min
(t =) 2.2 × 106 / 250 or (600 × 60) / 250 or 1 × 600 / 250

(c) Any two valid point(s):

• less noise or no noise
• less; or no air / gaseous pollution (from the bicycle)
• (the bicycle) uses no / less fossil fuel
• does not contribute to greenhouse effect or does not release CO2

Q262 • May/June 2022 Paper 42 Question 9 — 4.7 Motors

(a)(i) 0 (A)

(a)(ii) (I = 12 / 2 =) 6.0 A

(I =) V / R in any form

(a)(iii) (I = 12 / 5 =) 2.4 A

(Rs = R1 + R2 = 2 + 3 =) 5 (Ω)

(b) (Rp = 6 / 5 =) 1.2 Ω
1 / Rp = 1 / R1 + 1 / R2 or (Rp =) R1 R2 /
1 / Rp = 1 / 2 + 1 / 3 or (Rp =) 2 × 3 / (2 + 3)

Q263 • Oct/Nov 2022 Paper 41 Question 7 4.7 Motors

(a)
X and Y / they become magnetised or they / strips have poles
strips in the centre have opposite (magnetic) poles or X and Y attract
X and Y touch / close switch / activate relay / complete circuit
(b)(i) 150 A
I = P / V in any form or 1.8 / 12 or 1800 / 12 or 1800 / 12 or 0.15
(b)(ii)
small(er) resistance mentioned
less thermal energy produced or wires do not melt or large current mentioned
(c)
flexible strips in series with motor
power supply in series with motor
expected answer:
W
flexible iron
S strips
X
12 V car
battery
Y
magnetising coil
plastic case Z
M
starter motor

Q264 • Feb/March 2023 Paper 42 Question 8 4.7 Motors

(a) downwards / into the page / anti-clockwise
(b) current, (magnetic) field, motion at right angles to each other
Key physics: magnetic field left right / N S current B positive negative
(c) (at vertical) the coil stops; or (at vertical) the coil overshoots and comes back
Any one valid point(s):
• (as the coil approaches vertical) the turning effect decreases
• (at vertical) the turning effect is zero
• (past vertical) the turning effect reverses / changes direction
(d) reverses the current
Any two valid point(s):
• (brushes) ensure current is maintained /
• coil rotates continuously / continues to move in the same direction
• (allows current to change direction) without wires getting tangled
• (reverses the current) every half turn / 180 degrees / or (reverses the current) when the coil is vertical / at right angles to the magnetic field

Q265 • May/June 2023 Paper 41 Question 2 4.7 Motors

(a)(i) (speed =) 38 m / s
a = Δv / Δt or (Δv =) aΔt or (Δv =) 7.2 × 5.3
(a)(ii) (resultant force =) 1 700 N
F = ma or (F =) ma or (F =) 240 × 7.2

(b)(i) (vector) has direction (as well as magnitude) or scalar does not have direction
(b)(ii) (velocity) changes (as direction of motion changes) or direction (of velocity) changes
(b)(iii) Any two valid point(s):
• because there is an acceleration / change in velocity / change in direction / change in momentum (which needs a resultant force)
• motorcyclist accelerates / changes momentum (because velocity / direction changes)
• (resultant) force is perpendicular to the motion (of the motorcycle) or a ∝ F

Q266 • May/June 2024 Paper 42 Question 2 4.7 Motors

(a) change in velocity per unit time or rate of change of velocity or (a =) Δv / Δt
(b)(i) 12 s
(Δt =) Δv / a or 13 / 1.1
(b)(ii) 570 000 N
F = ma or (F =) ma or (F =) 520 000 × 1.1
(b)(iii) (additional force is needed to overcome) friction or air resistance or drag

Q267 • Oct/Nov 2024 Paper 42 Question 7 4.7 Motors

(a) (e.m.f. is the electrical) work done (by a source in) moving a unit charge around a (complete) circuit
(electrical) work done moving a charge
(b) 230 V
(c) 9.2 A
R = V / I or (I =) V / R or 230 / 25
(d) (ammeter reading =) {1.6 + candidate's answer to 7(c)} A
Sum of currents into a junction = sum of currents out of junction or total current is sum of the current in the branches

Q268 • Feb/March 2025 Paper 42 Question 8 4.7 Motors

(a) (direction of force) down(wards)
magnetic field direction, current direction and force are mutually perpendicular
Key physics: magnetic field N S / left right current flows positive negative anticlockwise into paper
(b) (direction of force) reverses / changes by 180°
(c) (J carbon brushes)
Any one valid point(s):
• connect cell / circuit to coil / wire / split ring(s) / commutator
• maintains (continuous) connection
• prevent wires from tangling (as motor rotates)
(K coil)
Any one valid point(s):
• rotates / turns
• conducts / has a current in it
(L axle)
Any one valid point(s):
• allows the coil to rotate / turn
• allows motor to turn / spin
(M split ring commutator)
Any one valid point(s):
• keeps motor turning in the same direction
• reverses the connections to the coil (every half -turn)
• prevents wires from tangling (as the motor rotates)

Q269 • May/June 2025 Paper 42 Question 3 4.7 Motors

(a) (momentum =) mass × velocity or (p =) mv
(b) (velocity =) 0.19 m / s and (direction) to the right

(momentum before collision) = {0.45 × 0.34} - {0.21 × 0.12} or 0.1278
(momentum after collision) = 0.66 × v
momentum before collision = momentum after collision
or {0.45 × 0.34} - {0.21 × 0.12} = 0.66 × v or 0.153 - 0.0252 = 0.66 × v
(c) (-) 0.12 N
FΔt = Δ{mv} or (F =) Δp ÷ (Δ)t or (F =) Δ{mv} ÷ (Δ)t or (F =) 0.26 ÷ 2.1

4.8 Transformers

Q270 • May/June 2021 Paper 41 Question 7 — 4.8 Transformers

(a)(i) Any three valid point(s):
- y-axis labelled e.m.f. and x-axis labelled time
- at least one cycle of a sinusoidal wave
- only two complete cycles of a sinusoidal wave
- constant amplitude and constant period for first two periods of a sinusoidal wave

(a)(ii) peak or trough or corresponding time labelled P
(a)(iii) (amplitude / maximum e.m.f.) increases
(e.m.f.) changes direction more often or greater frequency
(b) alternating current in primary coil
alternating / changing magnetic field
or magnetic field cuts secondary coil (continuously)
(alternating) e.m.f. induced in the secondary coil
(c) smaller current (and same resistance when the power is transmitted and an equal rate)
less thermal energy loss / produced (in cables)

Q271 • May/June 2021 Paper 43 Question 9 — 4.8 Transformers

(a) (NS =) 24 000
NS = NP × VS /VP or 50 × 110 × 103 / 230 in any form
(b) labelled diagram showing:
- (soft)-iron core
- copper coils
- fewer coils on secondary than primary

(c) alternating voltage in primary
alternating / varying / changing magnetic field (in iron core)
voltage is induced in the secondary coil

Q272 • Oct/Nov 2021 Paper 43 Question 10 — 4.8 Transformers

(a)(i) 6.0 V
(VS =) NSVP / NP in any form or (VS =) (25 × 120) / 500
(a)(ii) 2.5 A or 2500 mA
(IS =) IP VP / VS in any form or (0.125 × 120) / 6.0
(b)(i) arrow right to left along loose part of wire or any other correct position
(b)(ii)
wire moves up
(reversing direction of the current) reverses the direction of force
(c) coil does not continue to rotate in the same direction

Q273 • Feb/March 2022 Paper 42 Question 8 — 4.8 Transformers

(a) Labelled diagram showing
- labelled Iron (core)
- two coils, labelled primary and secondary, (around core and connected to separate circuits)

• Fewer turns on labelled secondary coil
(b) (Primary voltage causes) an alternating current in primary coil
produces a changing magnetic field
(changing) field induces pd / e.m.f. (in secondary coil)
(c) 0.026
(Ns / Np =) Vs / Vp in any form

Q274 • May/June 2022 Paper 43 Question 9 — 4.8 Transformers

(a)(i) Any four valid point(s):
• needle oscillates (as magnet moves up and down)
• coil cuts magnetic field / magnetic field changes (as magnet moves)
• changing (magnetic) field induces voltage/current
• induced voltage/current opposes the motion/change causing it
• force, magnetic field and induced current are mutually perpendicular
(a)(ii) larger (maximum) deflection
(b) 2300
25 000 × 36 000
(NP =) VP NS / VS in any form or (NP =)
400 000

Q275 • Oct/Nov 2022 Paper 43 Question 11 — 4.8 Transformers

(a) at least one line on the left and one line on the right, outside coil
curved back over the top and under the base of the coil, towards the central core of the coil
at least two (straight vertical) lines inside coil
direction of arrow correct on at least one line and none wrong
(b) 910
NP / NS = VP / VS or (NS =) (VS / VP) × NP
000 × 33 000 11000 × 33
or (NS =) or (NS =)
400 000

Q276 • May/June 2023 Paper 41 Question 8 — 4.8 Transformers

(a) Any four valid point(s):
• alternating current in (primary coil)
• (current in primary generates) changing magnetic field
• iron core concentrates (magnetic) field or iron core transfers (magnetic) field (to secondary coil)
• secondary coil is in alternating / changing (magnetic) field or secondary coil cuts (magnetic) field
• e.m.f. induced (in secondary coil)
(b)(i) (number of turns =) 3000
Np / Ns = Vp / Vs or (Np =) Ns Vp / Vs or (Np =) 450 × 220 000 / 33 000
(b)(ii) (current =) 350 A
P = IV or (I =) P / V or (I =) 7.7 × 107 / 220 000
(I =) 3.5 × 10N

Q277 • May/June 2023 Paper 43 Question 7 — 4.8 Transformers

(a) (Ip =) 0.14 A
Vp Ip = Vs Is or (Ip =) Vs Is / Vp or (Ip =) 12 × 2.5 / 220
(Ip =) 12 × 2.5 / 220
(b) (turns ratio =) 18 (: 1)
Np / Ns = Vp / Vs or (Np / Ns =) Vp / Vs or (turns ratio =) 220 / 12

Q278 • May/June 2024 Paper 42 Question 7 4.8 Transformers

(a) (end of) one piece of steel brought close to (the end of) another piece

look to see if there is repulsion/attraction and test between different ends/poles

Any two valid point(s):

• repeat a valid test between the other pieces

• only magnets repel each other or the pieces that repel are magnets

• attractions at both ends indicates one of them is unmagnetised

or the piece that only attracts is unmagnetised

or the piece that does not repel (at both ends) is unmagnetised

(b)(i) 0.19 A

Ip Vp = Is Vs or (Ip =) Is Vs / Vp or (Ip =) 45 / 240 or Ip Vp = 45 or power in primary = power in secondary

(Ip=) 45 / 240

(b)(ii) labelled diagram showing:

• (soft) iron core

• copper (coils)

• primary and secondary (coils) labelled and fewer coils on secondary than on primary

Q279 • Oct/Nov 2024 Paper 42 Question 9 4.8 Transformers

(a) Any two valid point(s):

1 (magnitude is) constantly changing

2 (magnetic field is) stronger closer to the coil of wire

3 (the magnetic field is) perpendicular to the a.c. current (producing it)

4 (the magnetic field) changes direction

(b)(i) Any one valid point(s):

• secondary coil is in changing / varying magnetic field

• secondary coil is in the magnetic field of primary coil

voltage is induced (in the secondary coil)

(b)(ii) Any one valid point(s):

• There is no iron core (to strengthen field)

• transformer usually has an iron core to connect the two coils

• some energy transferred to thermal energy (in phone / surroundings)

(c)(i) 6000 s

E = VIt or (t =) E / IV or 4.5 × 104 / [12 × 0.63]

(c)(ii) (Q =) 38 C

I = Q / t or (Q=) It or 0.63 × 60

Q280 • May/June 2025 Paper 41 Question 8 4.8 Transformers

(a)(i) (A is carbon) brushes

(B is) slip rings

(a)(ii) strengthens the magnetic field (of the magnet)

(b) graph is sinusoidal with positive and negative e.m.f.

graph shows minimum of one cycle completed in 0.5 s

e.m.f. is a maximum at 0.00 s, curves to a minimum at 0.25 s and curves to maximum at 0.50 s or

e.m.f. is a minimum at 0.00 s and curves to a maximum at 0.25 s and curves to a minimum at 0.50 s

braille: e.m.f. starts at time zero at either maximum or minimum value

(c)(i) either:

less power loss (for same power transmission)

(because) P = I2R

low current (for same power transmission)

(allows) thinner / cheaper cables

Any one valid point(s):

• less power loss (for same power transmission)

• P = I2R
• low current (for same power transmission)
• thinner / cheaper cables

(c)(ii) 5400

Vp Np NpVs
= or (Ns =) or
Vs Ns Vp
450 × 300 000
(Ns =)
25 000

Q281 • May/June 2025 Paper 43 Question 9 4.8 Transformers

(a)(i) Any three valid point(s):

• (current in the primary coil generates a) changing magnetic field (in primary coil)
• (iron) core transfers the magnetic field (to the secondary coil)
• secondary coil cuts the magnetic field / secondary coil is in (changing) magnetic field
• an e.m.f. is induced (in the secondary coil)
• (induced) current changes direction because the magnetic field changes direction

(a)(ii) 240 V

Vs ÷ Vp = Ns ÷ Np or (Vs =) 12 × 20(÷ 1)

(b) 0.16 A

P = I2R or I2 = P ÷ R or I2 = 1.25 × 10-3 ÷ 0.05(0) or I2 = 0.025

(c) Any two valid point(s):

• less power / heating / energy losses
• thinner / cheaper cables
• pylons further apart / fewer pylons
• transfer energy over long(er) distance

4.9 Digital electronics and logic gates

Q282 • Feb/March 2021 Paper 42 Question 9 4.9 Digital electronics and logic gates

(a) I/P I/P O/P

0 0
0 1
1 0
1 1
I/P I/P O/P
0 0 0
0 1 1
1 0 1
1 1 1

(b) two inputs to curved face, sharp end with small circle and one output

(c)(i) 1

(c)(ii) and

input 1 and 1 gives output 1
any 0 input gives 0 output

Q283 • May/June 2021 Paper 43 Question 8 4.9 Digital electronics and logic gates

(a) digital signal only two states - low or high or 0 or 1

analogue signal any value

(b)

correct symbol for NOR gate
(c)(i) and
(c)(ii) rows 1, 2, 5, 6 all 1
rows 3, 4, 7, 8 all 0

Q284 • Oct/Nov 2021 Paper 42 Question 10
4.9 Digital electronics and logic gates

(a) or (gate)
(b) 0
(c) prevents electrocution; or metal case cannot become live
(if) live wire touches metal case
(d)(i) if current too high
fuse melts
(d)(ii) 13 A (circled)
fuse rating/value above but near (to) normal operating current/ 10 A
fuse rating/value slightly higher (than) normal operating current /10A

Q285 • Feb/March 2022 Paper 42 Question 9
4.9 Digital electronics and logic gates

(a)(i) (A signal that has one of) two possible states
(a)(ii)
A B C D
0 0 0 1
0 1 0 1
1 0 0 1
1 1 1 0
C all correct
D all correct
(b) NAND

Q286 • May/June 2022 Paper 41 Question 9
4.9 Digital electronics and logic gates

(a) digital (signal) consists of 1(s) and 0(s) / high value and low
analogue (signal) is (continuously) variable (in magnitude)
(b)
NOR (gate) and
(c)
A Q
B
(i.e. NOR gate symbol with two inputs joined seen)

Q287 • Oct/Nov 2022 Paper 42 Question 9
4.9 Digital electronics and logic gates

(a)(i)
(a)(ii)
(b)(i)
I1 I2 Z O
0 0 1
0 1 1
1 0 0
1 1 0
all Z correct
all O correct
(b)(ii) NOT
(c)
or gate / box labelled or

NAND gate / box labelled NAND
or gate with inputs I1 and I2 labelled and NAND gate with inputs I1 and I2 labelled